

O(log n) Time Complexity

O(log n) means the algorithm reduces the problem size by half (or a constant factor) at each step.

Examl Binary Search

```
def binary_search(arr, target):
    left, right = 0, len(arr) - 1

    while left <= right:
        mid = (left + right) // 2

        if arr[mid] == target:
            return True
        elif arr[mid] < target:
            left = mid + 1
        else:
            right = mid - 1

    return False
```

Cuts the array into half

So steps $\approx \log_2(n)$

Dividing by 2 repeatedly

```
n = 100

while n > 1:
    n = n // 2
    print(n)
```